

The gating properties of *Drosophila* NMJ glutamate receptors and their dependence on Neto

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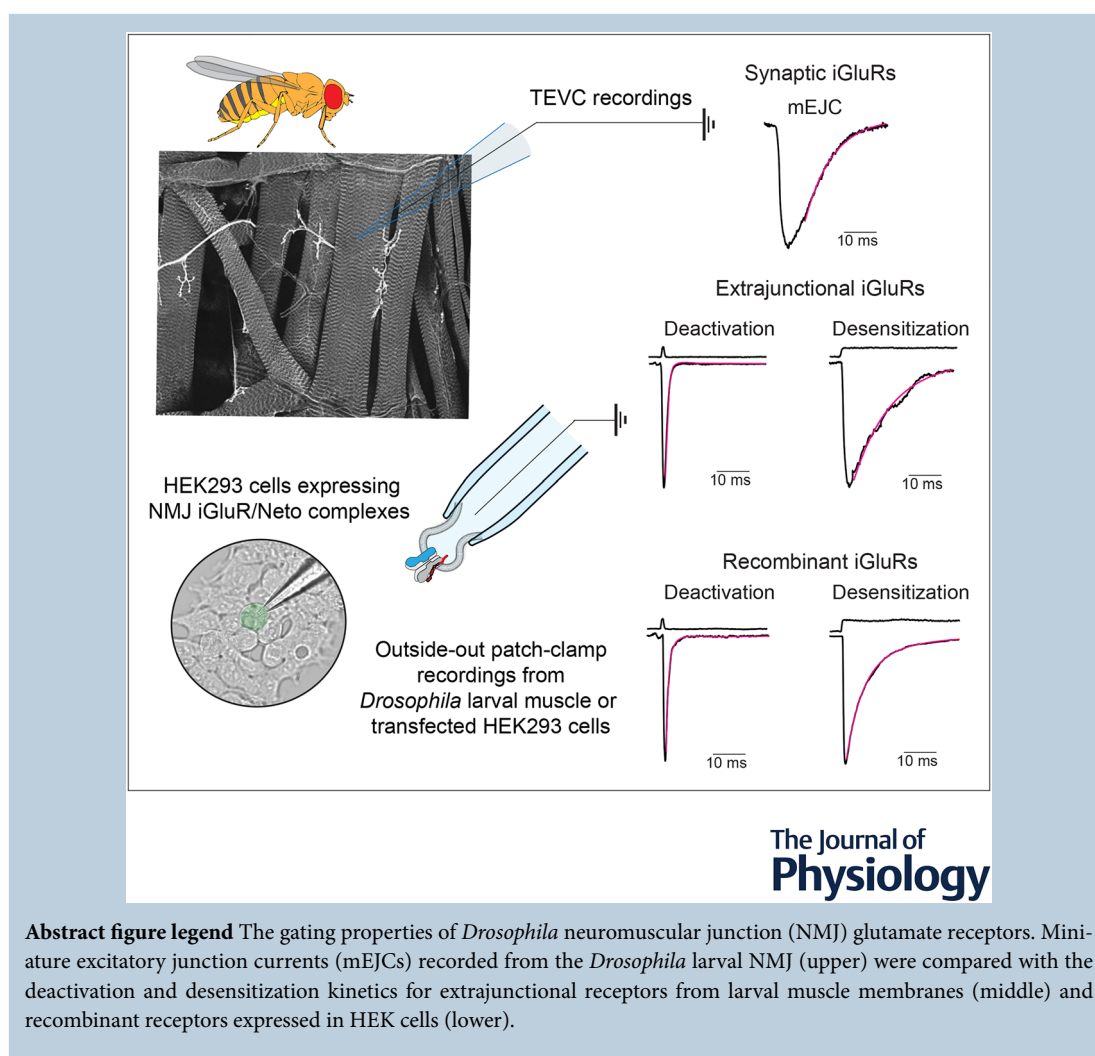
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Abstract The *Drosophila* neuromuscular junction (NMJ) is a powerful genetic system that has revealed numerous conserved mechanisms for synapse development and homeostasis. The fly NMJ uses glutamate as the excitatory neurotransmitter and relies on kainate-type glutamate receptors and their auxiliary protein Neto for synapse assembly and function. However, despite decades of study, the reconstitution of NMJ glutamate receptors using heterologous systems has been achieved only recently, and there are no reports on the gating properties for the recombinant receptors. Here, using outside-out, patch clamp recordings and fast ligand application, we examine for the first time the biophysical properties of native type-A and type-B NMJ receptors in complexes with either Neto- α or Neto- β and compare them with recombinant receptors expressed in HEK293T cells. We found that type-A and type-B receptors have strikingly different gating properties that are further modulated by Neto- α and Neto- β . We captured single-channel events and revealed major differences between type-A and type-B receptors and also between Neto splice variants. Surprisingly, we found that deactivation is extremely fast and that the decay of synaptic currents resembles the rate of ionotropic glutamate receptor (iGluR) desensitization. The functional analyses of recombinant iGluRs that we report here should greatly facilitate the interpretation of compound *in vivo* phenotypes of mutant animals.

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Key points

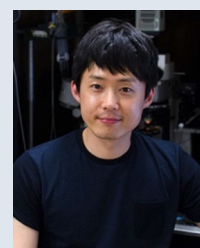
- We report the reconstitution of *Drosophila* neuromuscular junction ionotropic glutamate receptors (iGluRs) with Neto splice forms.
- Using outside-out patches and fast ligand application, we examine the deactivation and desensitization of the four iGluR/Neto complexes found *in vivo*.
- Expression of functional channels is absolutely dependent on Neto.
- Single-channel recordings revealed different lifetimes for different receptor complexes.
- The decay of synaptic currents is controlled by desensitization.

Introduction

Ionotropic glutamate receptors (iGluRs) mediate fast excitatory synaptic signalling throughout the vertebrate CNS and also at the neuromuscular junction (NMJ) of insects and crustaceans (Jan & Jan 1976; Takeuchi & Takeuchi 1964; Traynelis et al. 2010; Yu et al. 2021).

iGluRs are tetrameric ion channels that achieve strikingly diverse biophysical properties by combining different iGluR subunits with a rich array of auxiliary subunits (Hansen et al. 2021; Jackson & Nicoll 2011). Auxiliary subunits bind to iGluRs at many stages of the receptor life cycle and modulate not only channel properties, but also the delivery of receptors to the cell surface, their sub-cellular distribution, synaptic recruitment and association

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with various postsynaptic density (PSD) scaffolds. In *Drosophila*, multiple NMJ iGluR subunits (DiAntonio 2006), and the auxiliary protein Neto (Neuropillin and Toll-like) are each essential for viability and for embryos hatching into the larval stages (Kim et al. 2012), indicating that Neto is required for channel function at the NMJ. Phylogenetic studies indicate that fly NMJ iGluRs belong to the kainate receptor (KAR) clade, which has been expanded in *Diptera* (Li et al. 2016). Because Neto appears to be a KAR-dedicated auxiliary subunit (Tomita & Castillo 2012; Zhang et al. 2009), it would be anticipated that *Drosophila* Neto modulates the gating of NMJ iGluRs. However, the biophysical properties of *Drosophila* iGluRs and their modulation by auxiliary proteins remain poorly understood.

In flies, as in humans, synapse strength and plasticity are determined by the interplay between expression of different postsynaptic iGluR subtypes (DiAntonio et al. 1999). At the fly NMJ, type-A and type-B iGluRs are assembled from four different subunits: one copy of either GluRIIA (type-A) or GluRIIB (type-B), plus one copy each of GluRIIC, GluRIID and GluRIIE (DiAntonio et al. 1999; Featherstone et al. 2005; Marrus et al. 2004; Petersen et al. 1997; Qin et al. 2005). The shared iGluR subunits and Neto are essential for viability and for iGluR synaptic recruitment (DiAntonio 2006; Kim et al. 2012). Work in multiple laboratories demonstrated that the trafficking and stabilization of type-A and type-B receptors at synaptic sites follow distinct pathways and depend on receptor gating properties (Chen et al. 2005; Liebl & Featherstone 2008; Marrus & DiAntonio 2004; Marrus et al. 2004; Parnas et al. 2001; Petzoldt et al. 2014; Ramos et al. 2015; Sulkowski et al. 2016), but studies on the kinetic properties of these receptors are surprisingly sparse. Using outside-out patches isolated from the muscle membrane of wild-type larvae, a desensitization time constant of 14.6 ms was reported, but with a large patch-to-patch variation from 7 to 20 ms (Heckmann & Dudel 1997). Recordings from outside-out patches of larvae expressing only type-A or type-B receptors revealed a striking difference in desensitization, with average decay time constants of 19 and 2 ms, respectively (DiAntonio et al. 1999). However, no measurements for the deactivation of type-A and type-B receptors are available, and the role of the obligatory auxiliary protein Neto on the gating properties of NMJ iGluRs has not been explored.

Neto proteins are single-pass transmembrane proteins with a highly conserved extracellular part containing two C1r/C1s–Uegf–BMP domains (known as CUB1 and CUB2) and a low-density lipoprotein class A motif (LDL_A) and variable intracellular domains. Recent cryo-electron microscopy structures reveal that vertebrate Neto2 accesses different interfaces of GluK2 homotetrameric receptors, making tight intersubunit connections via CUB1/GluK2-ATD and LBD domains and intrasubunit

interactions within the transmembrane helices (He et al. 2021). In addition to the conserved protein interaction domains, *Drosophila* Neto has multiple intracellular putative docking motifs and phosphorylation sites, suggesting that Neto modulates the iGluR synaptic recruitment and function through dynamic interactions with other PSD components (Han et al. 2020; Kim et al. 2015; Ramos et al. 2015; Sulkowski et al. 2016). The *Drosophila* *neto* gene codes for two isoforms (Neto- α and Neto- β) with completely different intracellular domains of 206 and 351 residues, generated by alternative splicing (Ramos et al. 2015). Either isoform can sustain organism viability, but they have different *in vivo* distributions and functional roles (Han et al. 2015, 2020; Kim et al. 2012, 2015; Ramos et al. 2015).

We previously reported the functional reconstitution of recombinant *Drosophila* NMJ iGluRs in *Xenopus* oocytes assayed using two-electrode voltage clamp recording (Han et al. 2015). These studies showed that, just as in flies, four different subunits, either GluRIIA or GluRIIB, plus GluRIIC, GluRIID and GluRIIE, are required for the robust surface expression of *Drosophila* NMJ iGluRs; complexes assembled from fewer than four subtypes do not reach the cell surface and remain trapped in secretory compartments. Our experiments also revealed that, in the absence of Neto, *Drosophila* NMJ iGluRs are not activated by glutamate even after application of the lectin concanavalin A (Con A) (Han et al. 2015), thus establishing that, with heterologous expression, we can recapitulate results obtained *in vivo* using fly genetics. However, in contrast to vertebrate iGluRs, for which the kinetics of activation, deactivation and desensitization have been studied in depth, with extensive characterization of the properties of different subunit combinations and auxiliary proteins (Hansen et al. 2021), comparable studies on recombinant *Drosophila* iGluRs have not been reported. This represents a big gap in the field of developmental neurobiology: the *Drosophila* NMJ is a powerful system for studying glutamatergic synapse assembly, development and homeostasis, and extensive genetic manipulations of wild-type and mutant NMJ iGluR subunits have contributed to our understanding of underlying molecular mechanisms. However, the gating behaviour of these *Drosophila* NMJ iGluR variants was not determined and was assumed to mimic that of their mammalian counterparts, despite dramatic differences in the ligand binding properties of vertebrate and *Drosophila* NMJ iGluRs (Han et al. 2015).

In the present study, we examined the gating properties of *Drosophila* NMJ iGluRs using outside-out patch recordings from HEK293T cells transfected with different combinations of NMJ iGluR subunits and Neto splice variants. In addition, we generated flies with genetically controlled subunit composition and recorded responses from outside-out patches obtained from *Drosophila* larval

muscle, as well as excitatory synaptic currents from their NMJ. We used rapid application of glutamate to examine the kinetics of activation, deactivation and desensitization, the effect of polyamine toxins, and the effect of the lectin Con A. We recorded single-channel events which revealed different lifetimes and switches to low conductance sub-states for the type-A and type-B receptors. In the absence of Neto, we did not observe single events or macroscopic currents for either receptor. We find that the two Neto isoforms differentially modulate the deactivation and desensitization of type-A and type-B receptors. Our study reveals that *Drosophila* Neto is not only required for channel function, but also increases the repertoire of channel properties.

Methods

Molecular constructs and expression of iGluRs in HEK293T cells

iGluR coding sequences placed between the Tolloid-related signal peptide (Serpe & O'Connor 2006) and an RGS_{H6} C-terminal tag, as previously described (Han et al. 2015), were subcloned into a cytomegalovirus expression vector, pRK5-IRES (Li et al. 2016). The pRK5-Neto constructs included full-length Neto- α and Neto- β , derived from GH11189 and RE42119, respectively (Han et al. 2015), Neto- Δ CTD (M1-D478) (Ramos et al. 2015) and PM-Neto- Δ CTD (R¹²³I and R¹²⁶T) (Kim et al. 2015). HEK293T cells were cultured in Dulbecco's modified Eagle's medium (Gibco, Waltham, MA, USA) with 10% fetal bovine serum, 1% glutamax and 1% penicillin-streptomycin solution at 37°C in a 95% oxygen and 5% carbon dioxide incubator. HEK293T cells (2×10^5 per mL) were plated on 5-mm diameter coverglass coated with bovine collagen (Nutragen, Kerala, India), attached for 24 h, and then transfected with pRK5-based constructs of iGluR subunits and Neto variants ($1 \mu\text{g}$ total DNA mL⁻¹ cells) using ViaFect transfection reagent (Promega, Madison, WI, USA). The cells were incubated at 37°C for 3 h, and then the temperature was decreased to 30°C. Outside-out, patch clamp recordings were performed 2–3 days after transfection at room temperature.

Electrophysiological techniques

Outside-out patch recordings from HEK293T cells and muscle 6 (segment A3), with fast solution exchange achieved using four-bore glass tubing mounted on a P245.30 piezoelectric stack driven by a P-270 HVA amplifier (Physik Instrumente, Karlsruhe, Germany), were performed at room temperature as previously described (Horning & Mayer 2004). Recordings were made with thin-wall borosilicate glass pipettes (resistance,

3–6 M Ω). The external solution contained (in mM) 145 NaCl, 5.4 KCl, 1.8 CaCl₂, 1 MgCl₂, 5 Hepes (pH 7.3, osmolarity, 295 mOsm L⁻¹), to which 10 mM L-glutamate was added. The internal solution contained (in mM) 110 CsCl, 10 CsF, 0.5 CaCl₂, 1 MgCl₂, 10 Hepes, 5 CsBAPTA and 20 Na₂ATP (pH 7.3). Open tip junction potentials recorded at the end of experiments typically had 10–90% rise times of <300 μs , and data from patches for which responses to 10 mM glutamate had rise times >400 μs were discarded. Concanavalin A (type IV), glutamate and philanthotoxin-343 were purchased from Sigma-Aldrich (St Louis, MO, USA). Single-channel activity from outside-out patches was recorded using a gap-free protocol with an Axopatch 200B amplifier (Axon Instruments, Scottsdale, AR, USA), digitized with a Digidata 1550 (Molecular Devices, San Jose, CA, USA), sampled at 20 kHz, low pass filtered at 5 kHz and collected using pClamp10.7 (Molecular Devices). Synaptic currents were recorded using a gap-free protocol with an Axoclamp 2B amplifier (Axon Instruments), digitized with a Digidata 1440A (Molecular Devices), sampled at 20 kHz, low pass filtered at 1 kHz and collected using pClamp10.7 (Molecular Devices).

Fly strains and *in vivo* recordings

Four different types of mutations were used to study the gating properties of *Drosophila* NMJ glutamate receptor channels by receptor subunit and Neto isoform composition: *GluRIIA*^{SP16} (Petersen et al. 1997), *GluRIIB*[Mi03631] (BDSC# 37066), *neto- α* ^{null} (Han et al. 2020) and *neto- β* ^{null} (Ramos et al. 2015). These *GluR* mutants were crossed with a deficiency covering both *GluRIIA* and *GluRIIB* loci, *Df(2L)cl^{h4}* (Petersen et al. 1997); similarly, *neto* alleles were crossed with a *neto*^{null} mutant (Kim et al. 2012). To study the gating properties of extrasynaptic receptor channels using outside-out patch recording as described above, wandering third instar larvae were dissected in ice-cold, calcium-free hemolymph-like HL-3 saline, then incubated with 30 μg mL⁻¹ collagenase type IV (Sigma-Aldrich) for 10 min, washed with calcium-free HL-3 saline and moved to the recording chamber. The calcium-free HL-3 saline contained (in mM): 70 NaCl, 5 KCl, 20 MgCl₂, 10 HCO₃, 5 trehalose, 115 sucrose and 5 Hepes (pH 7.2). To examine the properties of synaptic receptor channels, two-electrode, voltage clamp recordings were performed on from muscle 6, segment A3 at room temperature as described previously (Qin et al. 2005). The recording solution was HL-3 with 0.5 mM CaCl₂. Intracellular electrodes (borosilicate glass capillaries of 1 mm in diameter) were filled with 3 M KCl with resistances ranging from 12 to 25 M Ω . Recordings were obtained from muscle cells with an initial membrane

potential between -50 and -70 mV, and input resistances of ≥ 4 M Ω . To record spontaneous miniature excitatory junction currents (mEJCs) the muscle cells were clamped to -80 mV.

Data analysis

To calculate deactivation and desensitization time constants, 50–100 representative responses were averaged and fit using a first-order exponential function for deactivation:

$$I(t) = I \exp(-t/\tau_{\text{off}})$$

and a double exponential function for desensitization:

$$I(t) = I_{\text{fast}} \exp(-t/\tau_{\text{fast}}) + I_{\text{slow}} \exp(-t/\tau_{\text{slow}})$$

where I_x is the peak current amplitude and τ_x is the corresponding decay time constant. To allow for comparison of decay times with published values fit with single exponential functions, weighted time constants τ_w were calculated using:

$$\tau_w = (I_{\text{fast}} / (I_{\text{fast}} + I_{\text{slow}})) * \tau_{\text{fast}} + (I_{\text{slow}} / (I_{\text{fast}} + I_{\text{slow}})) * \tau_{\text{slow}}$$

A non-stationary analysis of variance of responses to 10 mM glutamate, applied for 1 or 100 ms to outside-out patches from HEK cells transfected with various iGluR/Neto receptor complexes, was performed using Mini Analysis, version 6.03 (Synaptosoft, Fort Lee, NJ, USA). Individual responses to glutamate were aligned using the rise-time as a reference. Responses were then divided into 100 bins; for each bin, the variance (σ^2) and mean current (I) were calculated, with the baseline variance calculated for a 40 ms period prior to the application of glutamate subtracted. The resulting data was then plotted using KaleidaGraph, version 5.04 (Synergy Software, Byfleet, UK) to examine the variance–mean current relationship and fit to:

$$\sigma^2 = iI - I^2/N$$

where σ^2 is the variance, i is the unitary current, I is the mean current and N is the number of channels.

To calculate mEJC mean amplitudes, 50–100 events from each muscle were measured and averaged using the Mini Analysis program (Synaptosoft). For this analysis, we excluded events with humps and non-exponential decays (Pawlu et al. 2004). Our results obtained using two-electrode, voltage clamp recording agree well with previously published data obtained using extracellular focal recording, which reported a mean decay time constant of 4.4 ± 1.6 ms ($n = 13$) (range 2.5–7.7 ms) (Heckmann & Dudel 1998). In addition, we manually digitized additional published records for mEJC decay

obtained using extracellular focal recording and fit them with single exponentials (Cooper et al. 1995; Dawson-Scully et al. 2007; Karunanithi et al. 2018; Stewart et al. 1994). We estimated a mean value for τ_{mEJC} of 4.37 ± 0.46 ms ($n = 6$) (range 3.7–4.9 ms), which again is similar to values recorded in the current experiments, indicating that our recordings were not distorted by poor space clamp.

Single-channel analysis

Single-channel recordings collected using pClamp10.7 were first exported in MATLAB format. Custom MATLAB scripts were implemented to (1) rename different experimental variables for compatibility with the idealization software and (2) extract 150 ms intervals from the recordings, including 25 ms before and 25 ms after the 100 ms glutamate application. The idealization process was performed using a Python-based (<https://www.python.org>) open source, single-channel analysis application, ASCAM (<https://github.com/AGPlested/AS>) (Baranovic et al. 2022). This included (1) baseline correction using an offset method within the intervals [0, 0.025] and [0.125, 0.150] seconds; (2) Gaussian filtering (2 kHz cutoff); and (3) idealization itself using the ‘Analysis > Idealize’ function in ASCAM, with a multiple threshold crossing algorithm. Short-duration events were excluded by choosing a resolution of 200 μ s. We first identified the fully open state for all single channels, then isolated two additional open substates by eye, which corresponded to one-half and three-quarters of the open state. We never observed one-quarter open substates. The table of idealized events was imported into pClamp, and the distribution of log binned dwell times fit with exponential functions by maximum likelihood using pClamp10.7. A custom R pipeline (available upon request) was used to evaluate burst duration and open dwells for different subconductance states and the transitions between different substates.

Results

Drosophila NMJ iGluRs form Neto-dependent rapidly desensitizing receptors

To facilitate cell surface expression of recombinant *Drosophila* NMJ iGluRs, we replaced endogenous signal peptides with an optimized sequence and added a C-terminal RGSH6 epitope to all iGluR constructs. Four receptor subunits (GluRIIC, GluRIID, GluRIIE and either GluRIIA or GluRIIB) were transiently transfected in HEK293T cells with or without Neto splice variants and incubated at 30°C for 3 days prior to outside-out patch recordings. Because these receptors have low

sensitivity to glutamate (Han et al. 2015; Heckmann et al. 1996), we examined their gating properties in response to rapid application of 10 mM glutamate. In the absence of Neto, we observed no response to glutamate in 79 patches of GluRIIA/C/D/E and 46 patches of GluRIIB/C/D/E (Fig. 1A and B). However, when type-A receptor complexes were co-transfected with *Drosophila* Neto variants (Fig. 1C), a large fraction of outside-out patches (78/102) yielded macroscopic currents in response to glutamate. A similar requirement for Neto was observed for type-B receptors. Rapid application of 10 mM glutamate to outside-out patches

from HEK cells transfected with various iGluR/Neto combinations revealed subunit composition-dependent gating properties and differential effects of Neto variants (Table 1). For example, for macroscopic currents recorded from type-A iGluR/Neto- α (A/ α) complexes, the 10%–90% rise time was $330 \pm 38.58 \mu\text{s}$ ($n = 8$), with a deactivation time constant (τ_{off}) 0.64 ± 0.2 ms, whereas type-A iGluR/Neto- β (A/ β) complexes had a similar rise time, $340 \pm 71.07 \mu\text{s}$ ($n = 8$), but 1.4-fold slower deactivation, τ_{off} 0.90 ± 0.29 ms (Fig. 1D). Longer applications of glutamate (100 ms) revealed rapid and profound desensitization of more than 98% for both

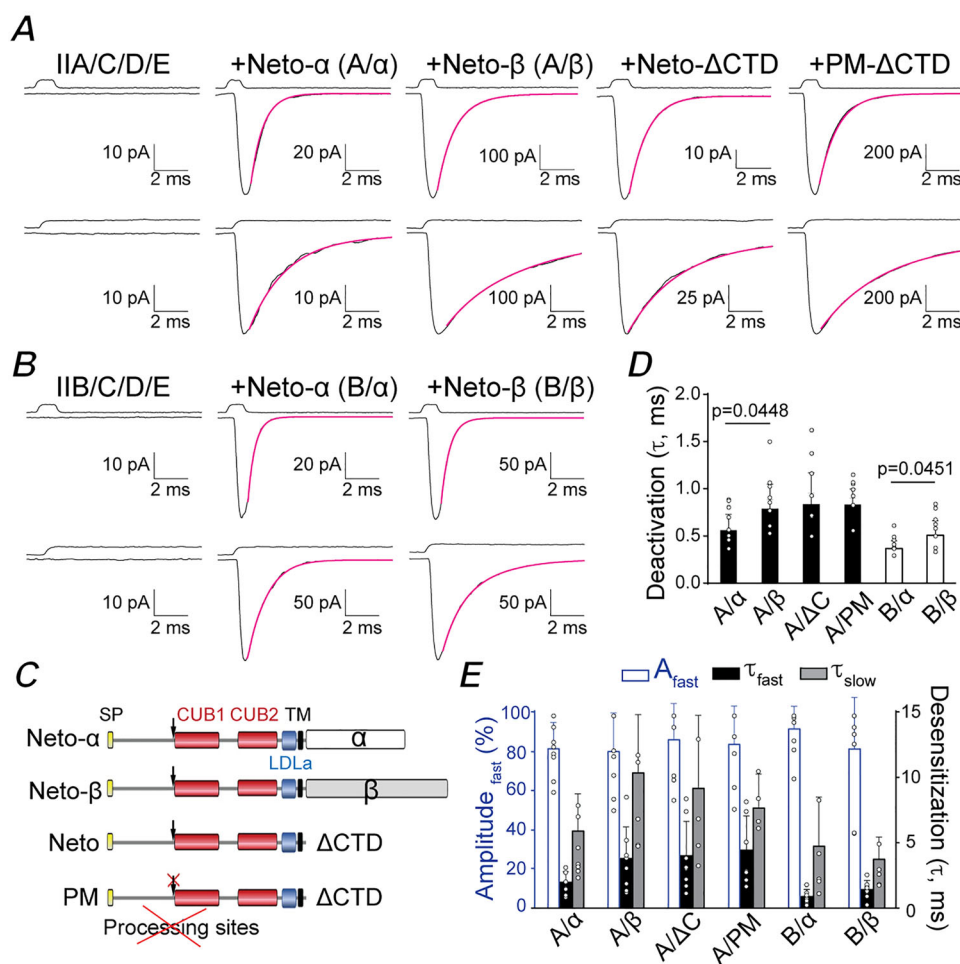


Figure 1. Differential modulation of *Drosophila* NMJ iGluRs by Neto isoforms

A and B, responses to 10 mM glutamate applied for 1 ms (upper traces) and 100 ms (lower traces) to outside-out patches from HEK293T cells transfected with GluRIIA/C/D/E (A) and GluRIIB/C/D/E (B) without Neto or in the presence of different Neto splice variants. Black lines show the average of 35–60 responses from one patch; magenta lines show the decay of the responses fitted with the sum of one (upper) or two (lower) exponential functions; open tip junction currents measured at the end of the experiments are shown at the top. The holding potential was -60 mV for all recordings. C, diagram of Neto variants utilized. *Drosophila* Neto isoforms are expressed as pre-proteins; their inhibitory pro-domain must be cleaved at conserved Furin processing sites (marked by arrows) before Neto can promote the formation of synaptic iGluR aggregates *in vivo*. PM denotes a processing mutant unable to shed the inhibitory pro-domain. D and E, summary graphs for deactivation (D) and desensitization time constants (E) for various iGluR/Neto complexes; the A_{fast} (%) component for desensitization is shown in blue. Data are represent the mean $\pm$ SD. [Colour figure can be viewed at wileyonlinelibrary.com]

Table 1. Upper segment: kinetic analysis for macroscopic currents evoked by 1 ms (deactivation) and 100 ms (desensitization) applications of glutamate recorded using either outside-out patches from HEK cells transfected with the indicated iGluR subunit combinations and Neto splice variants (recombinant) or patches obtained from muscle 6 for larvae with a defined subunit composition (extrajunctional); lower segment: kinetic analysis for mEJCs recorded using two-electrode voltage clamp for larvae with a defined subunit composition

	Cell type	Receptor type	Deactivation time			Desensitization time				N
			Rise time (ms)	τ (ms)	A (pA)	τ_{fast} (ms)	τ_{slow} (ms)	A_{fast} (%)	τ_w (ms)	
I/A/C/D/E+Neto- α	HEK293	Reconstituted	0.33 $\pm$ 0.04	0.64 $\pm$ 0.20	65.48 $\pm$ 86.56	2.01 $\pm$ 0.78	5.92 $\pm$ 2.83	81.49 $\pm$ 13.05	2.65 $\pm$ 0.98	8
I/A/C/D/E+Neto- β	HEK293	Reconstituted	0.34 $\pm$ 0.07	0.90 $\pm$ 0.29	221.99 $\pm$ 307.77	3.83 $\pm$ 2.37	10.37 $\pm$ 4.42	80.02 $\pm$ 19.36	5.18 $\pm$ 1.99	8
I/A/C/D/E+Neto- Δ CTD	HEK293	Reconstituted	0.34 $\pm$ 0.04	0.96 $\pm$ 0.39	72.88 $\pm$ 90.72	4.05 $\pm$ 2.60	9.19 $\pm$ 5.56	85.96 $\pm$ 18.23	4.72 $\pm$ 2.29	8
I/A/C/D/E+PM-Neto- Δ CTD	HEK293	Reconstituted	0.33 $\pm$ 0.03	0.95 $\pm$ 0.20	151.54 $\pm$ 317.69	4.47 $\pm$ 2.57	7.67 $\pm$ 2.57	83.70 $\pm$ 19.29	5.20 $\pm$ 2.00	7
I/B/C/D/E+Neto- α	HEK293	Reconstituted	0.27 $\pm$ 0.04	0.42 $\pm$ 0.10	52.40 $\pm$ 41.08	0.93 $\pm$ 0.50	4.77 $\pm$ 3.73	92.96 $\pm$ 11.99	1.14 $\pm$ 0.43	8
I/B/C/D/E+Neto- β	HEK293	Reconstituted	0.29 $\pm$ 0.04	0.58 $\pm$ 0.18	93.15 $\pm$ 103.14	1.45 $\pm$ 0.66	3.77 $\pm$ 1.68	81.32 $\pm$ 25.93	1.83 $\pm$ 0.42	8
I/A/C/D/E+Neto- α	Muscle 6	Extrajunctional	0.32 $\pm$ 0.06	1.57 $\pm$ 0.60	14.49 $\pm$ 10.95	0.89 $\pm$ 0.38	6.46 $\pm$ 2.19	74.61 $\pm$ 10.14	4.60 $\pm$ 2.14	5
I/A/C/D/E+Neto- β	Muscle 6	Extrajunctional	0.34 $\pm$ 0.04	1.60 $\pm$ 0.64	32.60 $\pm$ 18.31	7.39 $\pm$ 5.14	20.97 $\pm$ 9.27	84.13 $\pm$ 9.68	9.38 $\pm$ 5.44	6
I/B/C/D/E+Neto- β	Muscle 6	Extrajunctional	0.31 $\pm$ 0.03	0.97 $\pm$ 0.54	7.73 $\pm$ 1.63	0.92 $\pm$ 0.53	6.23 $\pm$ 3.49	69.67 $\pm$ 14.55	2.74 $\pm$ 1.73	5
	Cell type	Receptor type	Decay time			N				
			Rise time (ms)	τ (ms)	A (nA)					
I/A/C/D/E+Neto- α	Muscle 6	Synaptic	1.47 $\pm$ 0.08	4.20 $\pm$ 1.18	0.48 $\pm$ 0.14	8				
I/A/C/D/E+Neto- β	Muscle 6	Synaptic	1.42 $\pm$ 0.20	6.50 $\pm$ 0.79	0.66 $\pm$ 0.08	7				
I/B/C/D/E+Neto- β	Muscle 6	Synaptic	1.47 $\pm$ 0.13	4.92 $\pm$ 0.88	0.42 $\pm$ 0.13	7				

complexes ($n = 7-8$), with the decay best fit by the sum of two exponential functions: A/α τ_{fast} 2.01 ± 0.78 ms, τ_{slow} 5.92 ± 2.83 ms, A_{fast} $81.49 \pm 13.05\%$; and A/β τ_{fast} 3.83 ± 2.37 ms, τ_{slow} 10.37 ± 4.42 ms, A_{fast} $80.02 \pm 19.36\%$ (Fig. 1E).

Differential modulation by Neto isoforms was also observed for type-B receptors (Fig. 1B). These receptors had 10%–90% rise times of 267 ± 38.41 μ s ($n = 8$) and 294 ± 40.22 μ s ($n = 8$) for B/α and B/β , respectively, with deactivation time constants of 0.42 ± 0.10 ms and 0.58 ± 0.18 ms (Fig. 1D). Similar to type-A iGluRs, type-B receptors in complexes with Neto- α also desensitized faster than for Neto- β , with B/α τ_{fast} 0.93 ± 0.50 ms, τ_{slow} 4.76 ± 3.73 ms, A_{fast} $92.96 \pm 11.99\%$ ($n = 8$); and B/β τ_{fast} 1.45 ± 0.66 ms, τ_{slow} 3.77 ± 1.68 ms, A_{fast} $81.32 \pm 25.93\%$ ($n = 8$) (Fig. 1E). Of note, the deactivation and desensitization time constants for type-A iGluR/Neto- Δ CTD complexes, τ_{off} 0.96 ± 0.39 ms; desensitization τ_{fast} 4.05 ± 2.60 ms, τ_{slow} 9.19 ± 5.56 ms, A_{fast} $85.96 \pm 18.23\%$ ($n = 8$), were comparable to type-A iGluR/Neto- β complexes but different from type-A iGluR/Neto- α (Fig. 1C–E). This indicates that a ‘minimal Neto’, which retains the highly conserved extracellular and transmembrane domains, but lacks any intracellular parts, is sufficient for NMJ iGluR function. These results also reveal that the cytoplasmic domain of Neto- α , but not of Neto- β , modulates the gating properties of NMJ iGluR receptors, increasing the rate of deactivation and desensitization. Similar results were obtained for type-A receptors in complex with PM-Neto- Δ CTD (Fig. 1C–E), a processing mutant variant that cannot shed its inhibitory pro-domain and fails to cluster iGluRs *in vivo* (Kim et al. 2015). This indicates that responses observed in outside-out patches are probably independent of receptor clustering. To allow comparison with prior studies on extrajunctional *Drosophila* NMJ iGluRs, for which desensitization time constants with single exponential fits of 17.5, 18.8 and 2.0 ms were reported for wild-type, type-A and type-B, respectively (DiAntonio et al. 1999), a weighted value (τ_w) was calculated (see Methods). We found that type-A recombinant NMJ iGluRs channels have faster desensitization compared to native wild-type and native type-A receptors, with weighted decay time constants τ_w A/α , 2.65 ± 0.98 ms ($n = 8$) and A/β , 5.18 ± 1.99 ms ($n = 8$), whereas the rate of desensitization of B/α , 1.14 ± 0.43 ms ($n = 8$) and B/β , 1.83 ± 0.42 ms ($n = 8$), respectively, is comparable to native type-B iGluRs at the NMJ. We did not perform twin pulse applications of glutamate to measure the kinetics of recovery from desensitization, but note that prior studies on wild-type receptors report rapid recovery, time constant 150 ms (Heckmann & Dudel 1997).

In many recordings from patches with macroscopic currents, we observed trial-to-trial fluctuations in amplitude for both short (1 ms) and long (100 ms)

applications of glutamate. Given the large number of channels in each patch ($\sim 10-50$), these fluctuations probably occur because the open probability (P_o) is relatively low. We examined this possibility using a non-stationary analysis of variance (Fig. 2A–D, quantified in 2E–F). In many patches, the open probability was too low to accurately estimate, range (0.1–0.6), consistent with a prior study that reported a value of 0.4, estimated from binomial analysis of data from a single patch (Heckmann et al. 1996). Estimates of the single-channel conductance obtained from non-stationary analysis of variance, 160 ± 51 pS, 164 ± 43 pS and 149 ± 87 pS for A/α , A/β and B/α , respectively, gave values comparable to those reported previously from single-channel recordings from extra-junctional NMJ iGluRs (Chang et al. 1994; DiAntonio et al. 1999). In many patches with macroscopic currents, we observed steps corresponding to the closure or desensitization of individual channels (Fig. 2A–D).

Modulation of *Drosophila* NMJ iGluR single-channel responses by Con A

The lectin Con A attenuates iGluR desensitization for a wide variety of species, including locust NMJ iGluRs (Mathers & Usherwood 1976), native vertebrate kainate receptors (Huettner 1990; Wong & Mayer 1993) and *AvGluR1* from the primitive eukaryote *Adineta vaga* (Lomash et al. 2013). A recent cryo-EM structure of vertebrate *GluK2* in complex with succinyl Con A reveals that Con A binds at the interface between the amino-terminal domain and ligand-binding domain, preventing the large conformational change that mediates desensitization (Meyerson et al. 2016), and stabilizing the receptor in a state which cannot desensitize (Gangwar et al. 2024). In prior work, which used *Xenopus* oocytes to study recombinant *Drosophila* NMJ iGluRs, whole cell responses to glutamate were only detectable after treatment with Con A. To study the effect of Con A on single-channel activity, we treated HEK cells transfected with iGluR/Neto combinations with 0.6 mg mL^{−1} Con A for 10 min, and then excised outside-out patches to record responses to 10 mM glutamate applied for 100 ms. We found that patches from HEK cells treated with Con A were less stable, and that it was more difficult to obtain giga-ohm seals, but in exceptional cases sufficient data was obtained for kinetic analysis. We found that by reducing the incubation time after transfection from 3 to 2 days, we could obtain patches where only single-channel currents could be detected (Fig. 3A and B). For control patches, the single-channel conductance at -60 mV measured using individual openings was A/α 175 ± 12 pS ($n = 14$), A/β 172 ± 11 pS ($n = 16$), B/α 173 ± 12 pS ($n = 10$) and B/β 169 ± 9 pS ($n = 11$), similar to estimates from non-stationary analysis of variance (Fig. 2). From

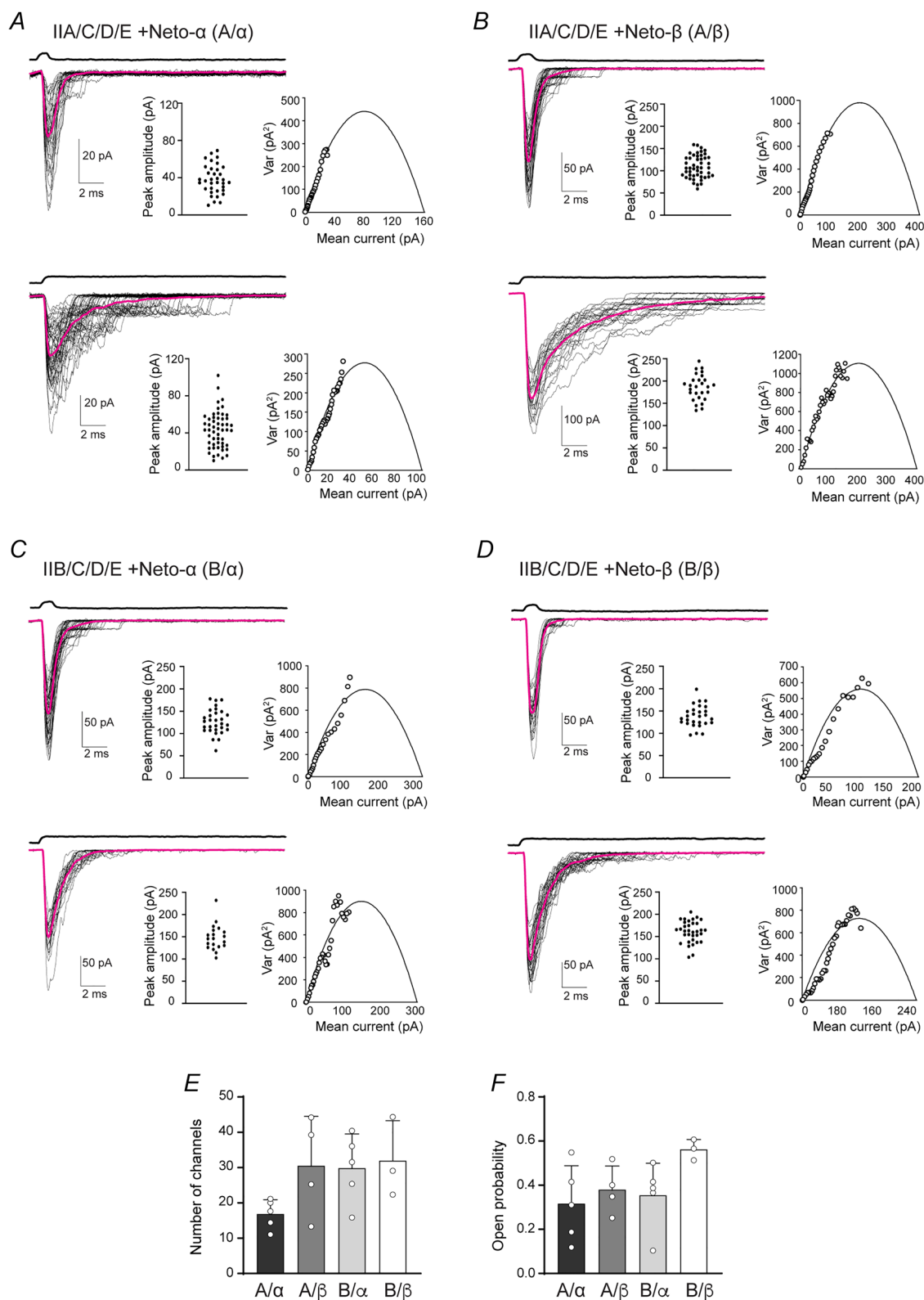


Figure 2. Non-stationary analysis of variance of responses to 10 mM glutamate applied for 1 ms (upper traces) and 100 ms (lower traces) to outside-out patches from HEK cells transfected with various iGluR/Neto receptor complexes as indicated

A–D, superimposed individual responses are shown in black; the average current in magenta; open tip junction currents measured at the end of the experiments are shown at the top. The holding potential was -60 mV for all

recordings. The insets in each panel show the peak amplitude for all trials (left) and the current-variance relationship (right) fit with the function $\sigma^2 = iI - I^2/N$, where σ^2 is the variance, i is the mean current, N is the number of channels. E , the mean number of channels and the open probability (F) determined by variance analyses of responses for three to five patches for different iGluR/Neto channel complexes as indicated, showing responses for individual patches and the mean $\pm$ SD. [Colour figure can be viewed at wileyonlinelibrary.com]

averages of 16–57 responses from single-channel patches we estimated desensitization time constants by fitting single exponentials: $\tau_{\text{des}} A/\alpha$ 2.04 ± 0.78 ms ($n = 6$), $\tau_{\text{des}} A/\beta$ 3.71 ± 1.12 ms ($n = 8$), $\tau_{\text{des}} B/\alpha$ 1.05 ms ($n = 2$) and $\tau_{\text{des}} B/\beta$ 1.54 ± 0.34 ms ($n = 3$); these values are comparable to those obtained for macroscopic currents recorded from multichannel patches (Fig. 1 and Table 1) with faster desensitization for type-B vs. type-A and for Neto- α vs. Neto- β .

In control patches not treated with Con A, single-channel events were observed only at the start of the application of glutamate (Fig. 3A and B), reflecting the rapid onset of desensitization observed for macroscopic responses (Fig. 1). By contrast, pretreatment with Con A dramatically increased single-channel activity, revealing major differences between type-A and type-B receptors and also between Neto splice variants. For A/ α , the response to glutamate showed a much slower onset desensitization, a time constant of 15.5 ms (range 10–26 ms, $n = 3$), which is six-fold slower vs. the value for control patches, time constant 2.65 ms, with very few openings 70 ms after the start of the application of glutamate (Fig. 3C). By contrast, for A/ β , B/ α and B/ β , there was a complete blockade of desensitization for some trials, whereas the average current revealed variable extents of desensitization from patch to patch (Fig. 3C and D). After treatment with Con A, single-channel activity for A/ α and especially A/ β consisted of bursts of long-duration openings, with few closures within a burst (Fig. 3C); by contrast, bursts for B/ α and B/ β were interrupted by brief closures at high frequency (Fig. 3D). Following termination of the application of glutamate, the average response after treatment with Con A showed tail currents with fast decays, time constant A/ α 1.9 ms $\pm$ 0.4 ms ($n = 4$), A/ β 3.8 ms $\pm$ 1.0 ms ($n = 5$), B/ α 0.9 ms $\pm$ 0.3 ms ($n = 4$) and B/ β 1.2 ms $\pm$ 0.1 ms ($n = 3$), suggesting that even after application of Con A glutamate dissociates rapidly following channel closure. However, these time constants are two- to four-fold slower than the rates of deactivation measured from control patches (Table 1), indicating that Con A might stabilize the open state.

The effects of Con A occurred without any change in single-channel conductance (Fig. 3E and G). Calculation of the charge transfer by integration of raw data in response to the application of glutamate for 100 ms revealed substantial differences between type-A and type-B receptors, and also between Neto- α vs. Neto- β . For control patches, the charge transfer was A/ α 0.37 ± 0.31

pC ($n = 4$), A/ β 0.47 ± 0.35 pC ($n = 8$), B/ α 0.12 pC ($n = 2$) and B/ β 0.14 ± 0.02 pC ($n = 3$), and there was a large increase for patches treated with Con A (Fig. 3F and H), with charge transfers of A/ α 3.33 ± 1.86 pC ($n = 5$), A/ β 5.14 ± 3.39 pC ($n = 6$), B/ α 1.34 ± 1.31 pC ($n = 5$) and B/ β 2.27 ± 2.79 pC ($n = 5$). Estimates of open probability were obtained from the analysis of all point amplitude histograms for pooled data from two to eight patches (Fig. 3I and J). Because the open state histograms were not well fit by single Gaussian functions, we calculated the open probability (P_o) by integration, which gave control values of 0.039, 0.044, 0.015 and 0.016 for A/ α , A/ β , B/ α and B/ β and 0.37, 0.55, 0.20 and 0.29 for patches treated with Con A (Fig. 3I and J).

Single-channel kinetics and subconductance states

Closer inspection of single-channel records for A/ α and A/ β complexes revealed bursts of openings, during which the current fluctuated between the main state and well-defined subconductance states with amplitudes of 75% and 50% (Fig. 4A and B); we did not observe openings to 25%, but it is possible these might occur with lower concentrations of glutamate. Using a cutoff time of 200 μ s, open time and burst duration histograms for A/ α and A/ β were well fit by the sum of two exponentials for both control patches, and for patches from HEK293T cells pretreated with Con A (Fig. 4C–F and Table 1). For A/ α , this analysis revealed a two- to four-fold increase in the duration of the slow component of the open time, 3.75 vs. 7.53 ms, and burst length, 3.12 vs. 11.53 ms, for control and Con A, respectively. For A/ β , the duration of the fast component of the burst length, 1.28 vs. 3.32 ms, and open time, 1.11 vs. 5.60 ms, also increased by 2.6- and five-fold, respectively for Con A treated patches. Notably, for type-A/ β , the duration of the slow component of the open time and burst length distribution increased even more dramatically, from control values of 4.51 and 4.38 ms to events of duration longer than 100 ms (Fig. 4D and F and Table 1), frequently with channel closure only after the termination of the application of glutamate (Fig. 3C).

Analysis of the event list generated by the single-channel idealization process revealed subtle differences between channels containing Neto- α vs. Neto- β (Fig. 5A–D). For example, A/ α spends more time in the 1/2 open substate than A/ β , which appears to spend a longer time in the 3/4 open substate. Application of Con A expanded these differences: A/ α channels visit the closed and 1/2 open subconductance states very

often, whereas A/ β channels rarely transition into these states and instead spend most of the time in the 3/4 open and full open substates. Because all the substates are well represented during a burst, we conclude that, upon desensitization block, A/ α channels transit between all subconductance states. By contrast, ~50% of A/ β channel bursts showed few substate transitions, whereas the other

50% had ~100 transitions per burst, suggesting modal gating. Because of the brief lifetime of openings observed for type-B receptors (Fig. 3B) we did not attempt a similar single-channel kinetic analysis, but note that inspection of the raw data suggests that the burst length of B/ α is shorter than for B/ β similar to the behaviour observed for A/ α and A/ β .

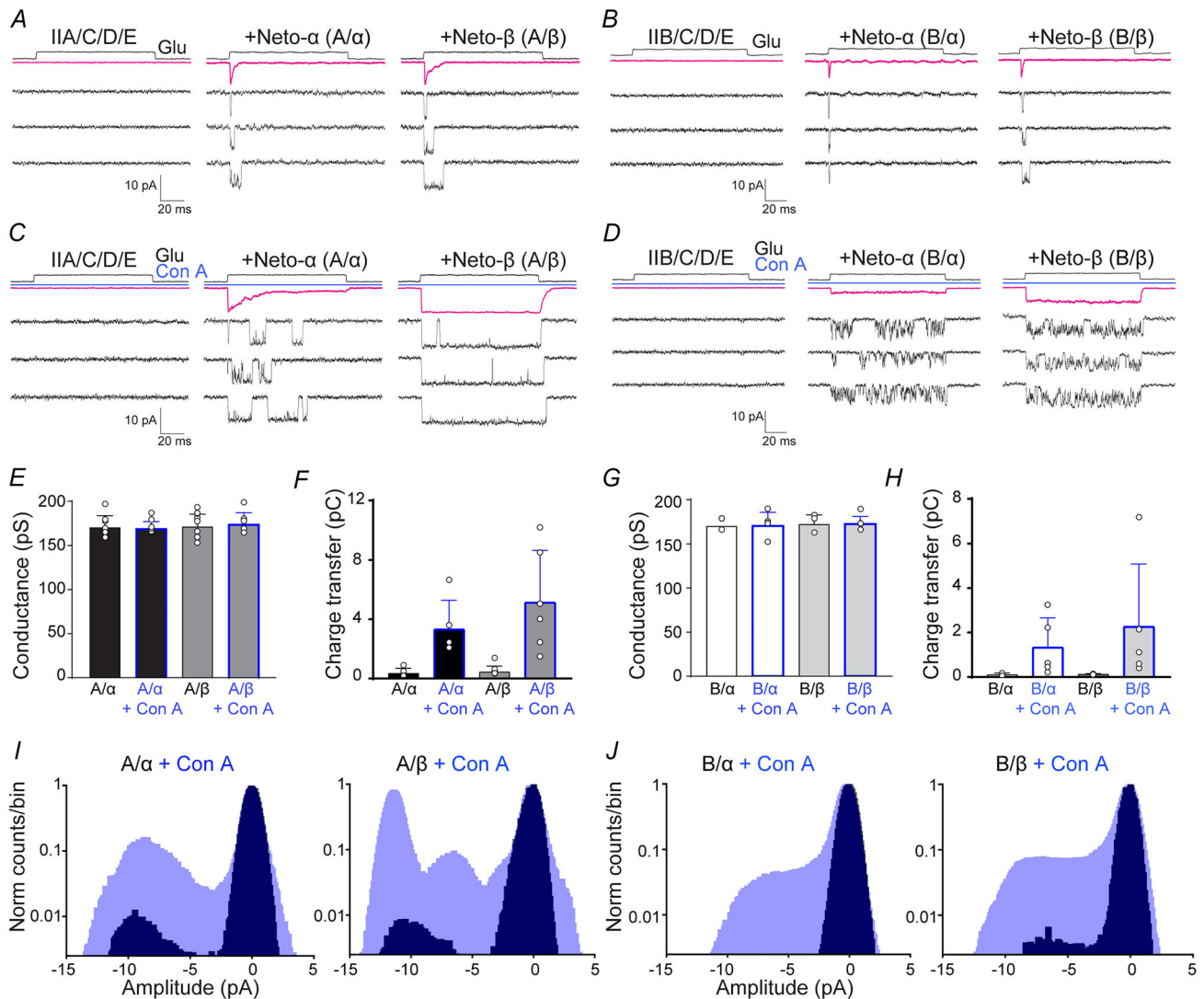


Figure 3. Concanavalin A attenuates desensitization and increases single-channel activity evoked by glutamate

A and C, responses for outside-out patches to 10 mM glutamate applied for 100 ms to HEK293T cells transfected with type-A receptors with or without Neto splice variants, as indicated, before (A) and after (C) treatment with 0.6 mg mL⁻¹ Con A for 10 min (blue line); average responses (top, magenta line) and three single-channel responses from one patch are shown for each channel variant. The same sequence is shown for type-B receptors (B and D). The holding potential was -60 mV for all recordings. Open tip junction currents measured at the end of the experiments are shown at the top. E and F, single-channel conductance at -60 mV and the charge transfer for responses to glutamate measured by integration of 35–55 trials shown in (A) and (C) for type-A receptors. G and H, showing the same analysis for type-B receptors. I and J, all points amplitude histograms plotted on a log scale before (black) and after (blue) treatment with Con A, normalized to the closed state peak at 0 pA. Note the shift in the peak profiles at -10 pA (open channels) after application of Con A and the asymmetric profile for the open state. Data represent the mean $\pm$ SD. [Colour figure can be viewed at wileyonlinelibrary.com]

Native NMJ iGluRs and synaptic currents

Differences between the properties of native and recombinant neurotransmitter receptors have been instrumental in the discovery of novel auxiliary proteins and synaptic modulators (Jackson & Nicoll 2011). *Drosophila* NMJ iGluR receptor channel subtypes are assembled from either of two alternative subunits, GluRIIA and GluRIIB, combined with two Neto isoforms. Null mutants are available for each of the four variants (DiAntonio et al. 1999; Han et al. 2020; Kim et al. 2015), permitting the *in vivo* isolation of individual receptor types with a single Neto isoform; this facilitates a direct comparison with recombinant iGluR/Neto channels expressed in HEK293T cells. We generated third-instar larvae with single copies of either the *GluRIIA* or *GluRIIB* genes, and either *neto-α* or *neto-β*. Mutants with one or

two copies of *GluRIIB* and *neto-α* (*neto-β^{null}*; *GluRIIA^{null}*) are embryonic lethal and thus could not be studied.

We recorded from outside-out patches obtained from the larval muscle membrane (muscle 6, abdominal segment 3) of animals with defined receptor complexes and compared the deactivation kinetics of responses to 1 ms applications of 10 mM glutamate with those of recombinant receptors of the same subunit composition expressed in HEK cells. We found that native, extrajunctional channels have slower deactivation kinetics than recombinant receptors (Figs 1, 6A and B and Table 1), *A/α* τ_{off} , 1.57 ± 0.60 ms in muscle patches vs. 0.64 ± 0.20 ms in HEK cells and *A/β* τ_{off} , 1.60 ± 0.64 ms vs. 0.90 ± 0.29 ms, ($n = 5$ or 6). A less pronounced but similar trend was observed for *B/β*: recombinant receptors had faster deactivation τ_{off} , 0.58 ± 0.18 ms, than

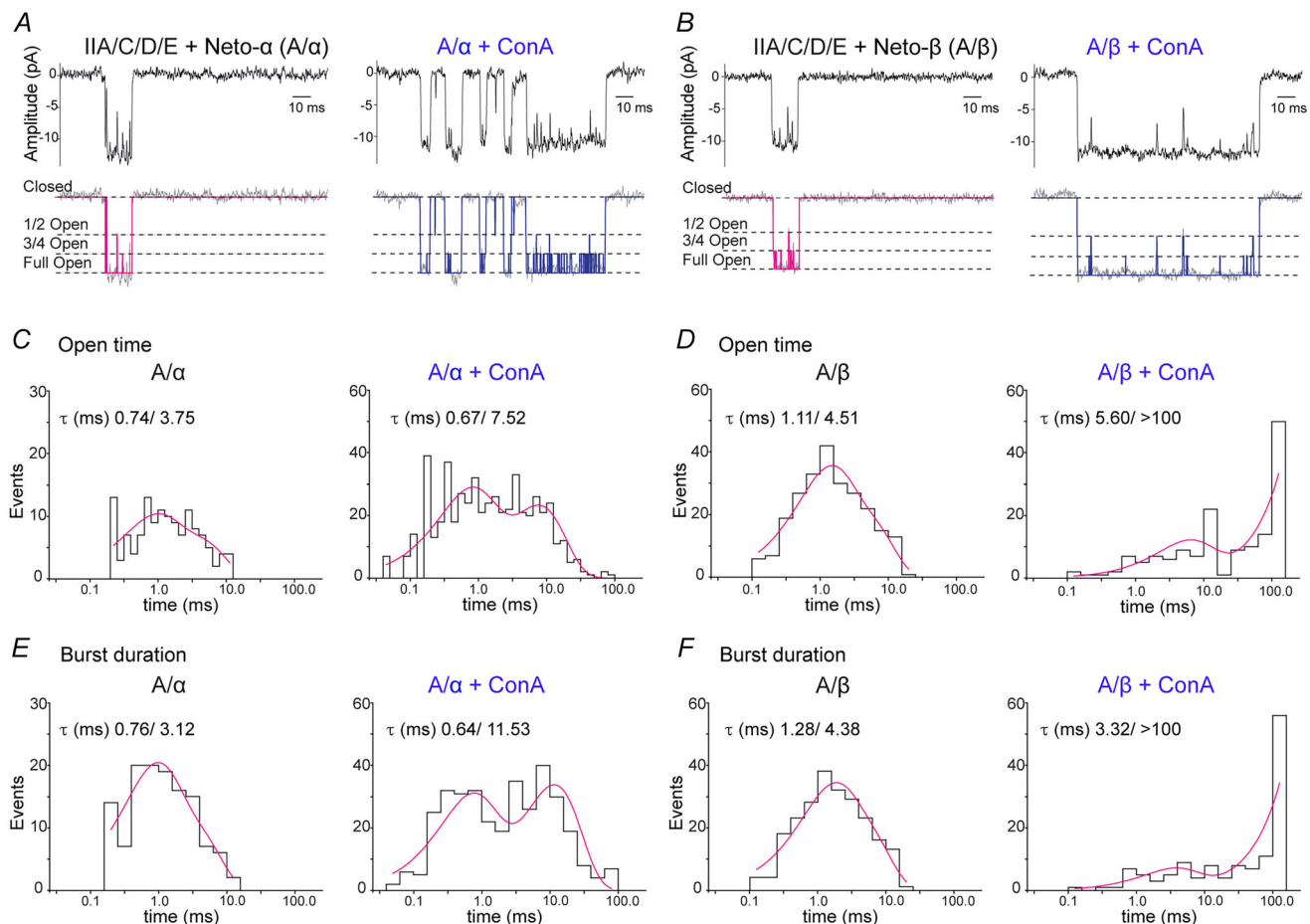


Figure 4. Analysis of single-channel kinetics for type-A receptors

A and B, filtered and idealized traces illustrating responses to 10 mM glutamate applied for 100 ms to outside-out patches from HEK293T cells transfected with *A/α* and *A/β* receptor complexes; left: control patches, right: patches obtained after treatment of HEK cells with 0.6 mg mL^{-1} Con A for 10 min; idealized traces, shown in magenta and blue, respectively, reveal the full open state and two subconductance states, half open and three-quarter open, as indicated by dotted lines. C and D, open time histograms for *A/α* or *A/β* receptor complexes for control patches (left) and patches after treatment with Con A (right) fit with the sum of two exponential functions as indicated. E and F, burst duration histograms for *A/α* and *A/β* for control patches (left) and after treatment with Con A (right). [Colour figure can be viewed at wileyonlinelibrary.com]

the native extrajunctional complexes τ_{off} , 0.97 ± 0.54 ms (Fig. 6C). By contrast, the single conductance γ : A/ α , 169.3 ± 6.58 pS; A/ β , 171.1 ± 8.60 pS; B/ β , 172.6 ± 7.73 pS for native receptors was not different from values obtained for recombinant receptors. We next recorded mEJCs at the larval NMJ in mutants with defined receptor complexes and estimated the decay time for synaptic currents fit with single exponential functions (Fig. 6D–F). The mEJC decay time constant was much slower than the deactivation time constant for both type-A and type-B extrajunctional receptors, τ_{mEJC} : A/ α , 4.20 ± 1.18 ms

($n = 5$); A/ β , 6.50 ± 0.79 ms ($n = 6$); B/ β , 4.92 ± 0.88 ms ($n = 5$) (Fig. 6G–I). The slow decay cannot be explained by asynchronous release at multiple junctional sites because we excluded mEJCs with slow rise times, or which had steps on their rising phase, or which had humps at their peak. Instead, our data suggest that either the clearance of glutamate from the synaptic cleft is slow compared to most vertebrate excitatory synapses, or that additional auxiliary membrane proteins and/or cytoplasmic modulatory proteins are present and impact the gating of synaptic receptors.

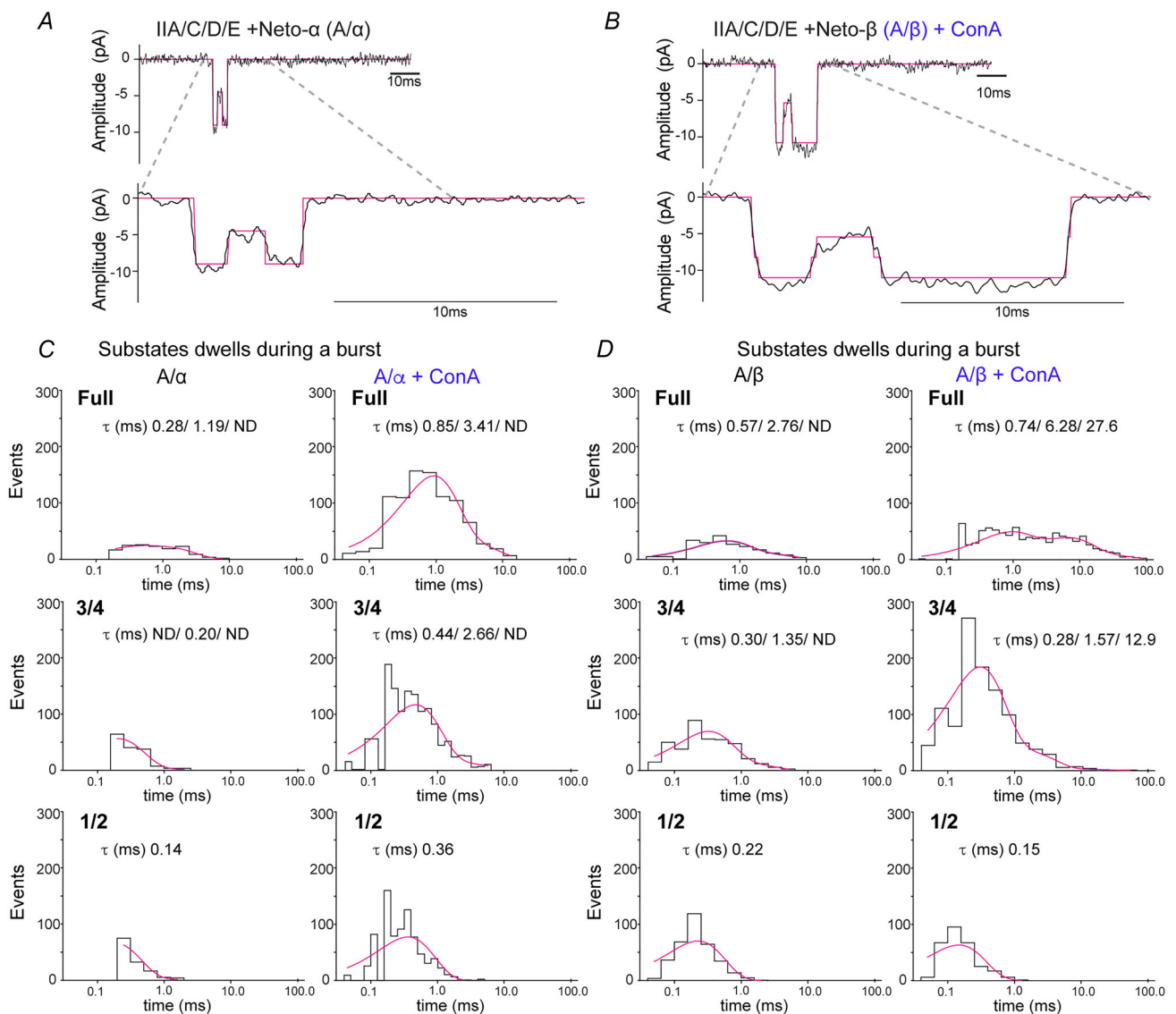


Figure 5. Modulation of substate activity by Con A

A and B, representative responses to 10 mM glutamate applied for 100 ms to outside-out patches from HEK cells transfected with A/ α and A/ β receptor complexes, as indicated. The idealized traces (in magenta) show transitions from the fully open state to the half level. C and D, dwell time histograms for substate occupancy during a burst for control patches (left), and for patches from HEK cells treated with Con A (right) for A/ α (C) and A/ β (D), fit with the sum of one to three exponentials of fast, intermediate and slow time constants, as indicated. ND, not detected. [Colour figure can be viewed at wileyonlinelibrary.com]

To compare the kinetics of mEJCs with the kinetics of desensitization we next recorded responses to 100 ms applications of 10 mM glutamate from native extrajunctional receptors (Fig. 6J–L, comparisons in 6M–O). This revealed that the desensitization time constant for A/ α , 4.60 ± 2.14 ms ($n = 5$) was almost identical to the mEJC decay time constant of 4.20 ms. The desensitization time constant for native A/ β , 9.4 ± 5.44 ms ($n = 6$) was 1.4-fold slower than the mEJC decay time constant of 6.5 ms. Curiously, the desensitization time constant for native B/ β , 2.74 ± 1.73 ms ($n = 5$) was 1.8-fold faster than the mEJC decay time constant of 4.9 ms.

Use-dependent blockade by external philanthotoxin (PhTx)

Ca²⁺-permeable vertebrate kainate receptors are blocked by extracellular PhTx (Bähring & Mayer 1998), a polyamine toxin derived from wasp venom (Eldefrawi et al.

1988). Because native *Drosophila* NMJ iGluRs have high Ca²⁺-permeability (Chang et al. 1994), we tested the effects of PhTx on multichannel outside-out patches at a concentration of 1 μ M, when recording the response of type-A and type-B receptors to glutamate (Fig. 7A and D). Before application of PhTx, the mean charge transfer in response to 100 ms applications of 10 mM glutamate was: A/ α , 4.08 ± 2.54 pC ($n = 5$); A/ β , 4.68 ± 3.95 pC ($n = 5$); B/ α , 0.17 ± 0.07 pC ($n = 4$); B/ β , 0.40 ± 0.27 pC ($n = 4$). Similar to the macroscopic currents shown in Fig. 2, we observed trial-to-trial amplitude variations for all channel combinations. Inhibition by PhTx developed slowly, τ_{onset} 78 s and 56 s for A/ β and B/ β , respectively (mean τ_{onset} 73.45 ± 13.14 s, $n = 5$ and 37.56 ± 10.21 s, $n = 4$) (Fig. 7A' and D'), but, at equilibrium, PhTx substantially reduced the charge transfer in response to a 100 ms application of glutamate to $1.6 \pm 0.5\%$ ($n = 5$) and $1.2 \pm 0.7\%$ ($n = 5$) of control, for type A/ α and A/ β channels respectively. For type-B channels, blockade at equilibrium was weaker, with the charge transfer reduced to only $16.7 \pm 5.1\%$ for

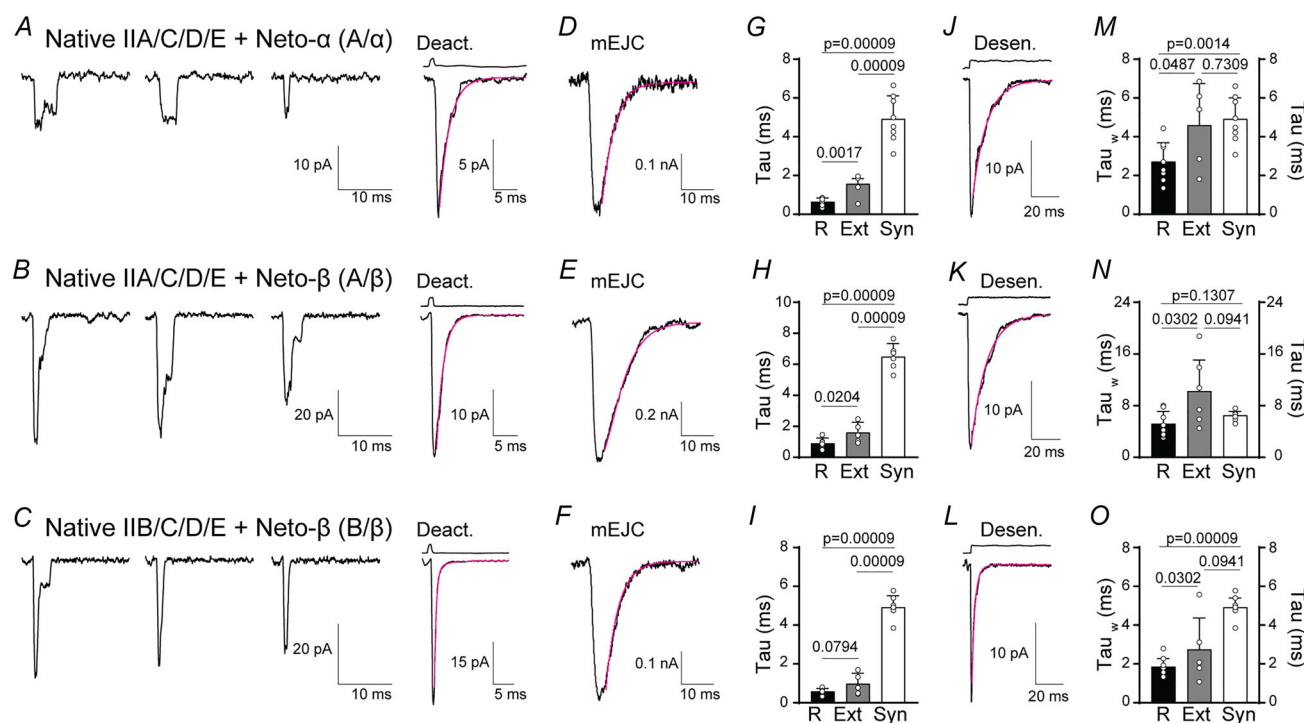
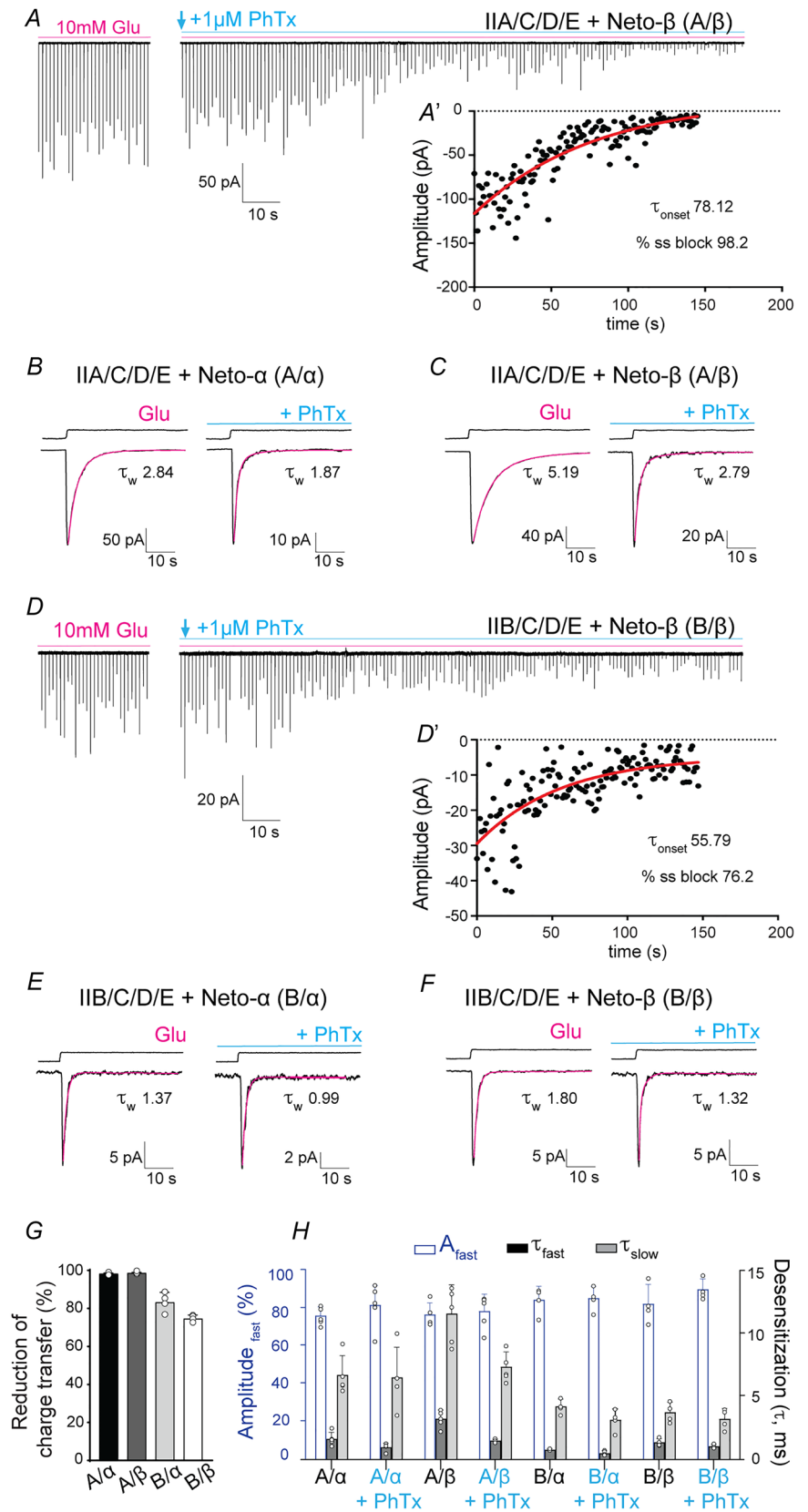


Figure 6. Distinct receptor properties observed for native and recombinant receptors

A–C, representative individual traces (left) and the average of 20–40 responses (right) to 10 mM glutamate applied for 1 ms to outside-out patches of native, extrajunctional receptors excised from body-wall muscles of third instar larvae with distinct iGluR/Neto complexes, as indicated. The magenta line shows the time course of deactivation fit by an exponential function; open tip junction currents measured at the end of the experiments are shown at the top. D–F, miniature EJCs recorded from larvae of the same genotypes fit with single exponential functions. G–I, comparisons of deactivation kinetics for recombinant receptors expressed in HEK cells (R), native, extrajunctional receptors from larval muscle membranes (Ext) and mEJCs (Syn). J–L, average of 20–40 responses to 10 mM glutamate applied for 100 ms to outside-out patches of native, extrajunctional receptors for the same genotypes; the magenta line shows fits of the sum of two exponential functions; open tip junction currents measured at the end of the experiments are shown at the top. M–O, comparisons of desensitization kinetics for recombinant receptors expressed in HEK cells (R), native, extrajunctional receptors from larval muscle membranes (Ext) and mEJCs (Syn). Data represent the mean $\pm$ SD. [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 7. Slow onset of block by external PhTx

A and D, representative traces for type-A and type-B receptors showing responses recorded from outside-out patches to 10 mM glutamate applied for 100 ms at an interval of 1 s before (magenta line) and after the application of 1 μ M PhTx (blue line). The amplitude variation is a result of differences in the number of channels activated from trial to trial. (A' and D') Data points indicating the amplitude of sequential responses to 10 mM glutamate applied for 100 ms at intervals of 1 s for A/ β and B/ β . Magenta lines show single exponential fits to the decay of the response to glutamate due to onset of block by 1 μ M PhTx. B, C, E and F, averages of 20 responses before PhTx or starting 5 s after the onset of PhTx application for A/ α (B), A/ β (C), B/ α (E) and B/ β (F). Magenta lines show fits of double exponential functions and reveal faster decay in the presence of PhTx. The holding potential was -60 mV; back lines above the response to glutamate show open tip potentials. G, the extent of block by PhTx at equilibrium, estimated from the change in charge transfer with respect to control, where a value of 100% indicates complete block, with values A/ α , $98.4 \pm 0.52\%$ ($n = 5$); A/ β , $98.8 \pm 0.67\%$ ($n = 5$); B/ α , $83.31 \pm 5.07\%$ ($n = 5$); B/ β , $74.70 \pm 1.72\%$ ($n = 5$). H, fits to the decay of the response to 100 ms applications of 10 mM glutamate (fit with the sum of two exponentials) before and after application of PhTx, showing mean values for %A_{fast}, τ_{fast} and τ_{slow} for the indicated combinations of type-A and type-B. Data represent the mean $\pm$ SD. [Colour figure can be viewed at wileyonlinelibrary.com]



B/ α ($n = 4$) and $25.3 \pm 1.7\%$ for B/ β ($n = 4$) compared to control (Fig. 7G). This difference in PhTx-induced blockade between type-A and type-B receptors resembles that observed in prior experiments using *Xenopus* oocytes to study blockade by the structurally related channel blocker argitoxin (Han et al. 2015).

The reduction in charge transfer by PhTx results from two effects. First, the number of channels that open in response to glutamate progressively decreases in the presence of PhTx (Fig. 7A and D). Second, the rate of decay of the response to glutamate, which, under control conditions results from the onset of desensitization, increases in the presence of PhTx; this is most probably the result of a combination open channel blockade by PhTx combined with the onset of desensitization. Indeed, in the presence of PhTx, the kinetics of decay were substantially faster, as estimated from the average response to 15–30 applications of glutamate recorded immediately before and after the start of the application of toxin, τ_w A/ α 2.84 ± 0.45 ms before and 1.87 ± 0.17 ms after PhTx ($n = 5$); τ_w B/ α 1.37 ± 0.27 ms vs. 0.99 ± 0.17 ($n = 4$); τ_w A/ β 5.19 ± 0.87 ms vs. 2.79 ± 0.51 ($n = 5$); and τ_w B/ β 1.80 ± 0.30 ms vs. 1.32 ± 0.14 ($n = 4$) (Fig. 7B, C, E and F, quantified in 7H). Most of this change was the result of an increase in the fast component of decay of the response to glutamate (Table 2). The slow rate of onset of blockade by PhTx probably reflects the low open channel probability (Fig. 2) convolved with the limited time that the channel spends in the PhTx sensitive open state because of the rapid onset of desensitization in response to glutamate.

Discussion

The fly NMJ has been used extensively for genetic analysis of synapse development and homeostasis. Because the trafficking and synaptic stabilization of NMJ iGluRs depends on receptor activity (Marrus & DiAntonio 2004; Petzoldt et al. 2014), our lack of an understanding of iGluR receptor properties and how they are regulated has been a decades long gap in the field. Here, we address this issue by measuring the function of recombinant *Drosophila* NMJ iGluRs expressed in HEK293T cells and comparing their gating with native extrajunctional and synaptic receptors. We find that the kinetic properties of type-A and type-B receptors are strikingly distinct from each other, with differential modulation by Neto isoforms adding further diversity to their kinetics. In the experiments reported here, we used outside-out patches with sub-millisecond solution exchange and found that, in the absence of Neto, *Drosophila* NMJ iGluRs are not activated by glutamate on a physiological time scale; this explains why *neto*^{null} mutants are embryonic lethal and why Neto is essential *in vivo* for NMJ functionality (Han et al. 2015; Kim et al. 2012). We cannot exclude the possibility that receptors

Table 2. Decay kinetics for responses to 100 ms applications of 10 mM glutamate recorded from the same outside patch before and in the continuous presence of 1 μ M PhTx

Cell type	Receptor type	Control					PhTx (+)					N
		Desensitization time					Desensitization time					
		τ_{fast} (ms)	τ_{slow} (ms)	A_{fast} (%)	τ_w (ms)	τ_{fast} (ms)	τ_{slow} (ms)	A_{fast} (%)	τ_w (ms)			
HEK293	Recombinant	1.69 ± 0.52	6.68 ± 1.49	76.06 ± 4.66	2.84 ± 0.45	1.04 ± 0.30	6.50 ± 2.33	81.79 ± 11.56	1.87 ± 0.17	5		
HEK293	Recombinant	3.27 ± 0.67	11.46 ± 2.21	76.69 ± 6.11	5.19 ± 0.87	1.55 ± 0.12	7.31 ± 1.15	78.48 ± 8.93	2.79 ± 0.51	5		
HEK293	Recombinant	0.86 ± 0.06	4.23 ± 0.56	84.47 ± 7.08	1.37 ± 0.27	0.59 ± 0.18	3.16 ± 0.86	85.30 ± 6.24	0.99 ± 0.17	4		
HEK293	Recombinant	1.41 ± 0.30	3.76 ± 0.78	82.33 ± 10.23	1.80 ± 0.30	1.12 ± 0.14	3.24 ± 0.82	89.98 ± 5.44	1.32 ± 0.14	4		

Responses were fit with the sum of two exponentials, revealing a 1.3- to 2.1-fold increase in the rate for the fast component of decay in the presence of PhTx.

Responses were fit with the sum of two exponentials, revealing a 1.3- to 2.1-fold increase in the rate for the fast component of decay in the presence of PhTx.

which traffic to the membrane in the absence of Neto desensitize so rapidly that the channel does not open, as found for the AMPA receptor S750D and E755A mutants (Horning & Mayer 2004; Sun et al. 2002).

Differences between vertebrate kainate receptors and the *Drosophila* NMJ iGluRs

It is important to emphasize that vertebrate kainate receptors function robustly in the absence of Neto. This is not true for postsynaptic *Drosophila* NMJ iGluRs (Figs 1 and 3) (Han et al. 2015). Studies on vertebrate kainate receptors reveal that coassembly with Neto alters receptor gating kinetics, slowing deactivation and desensitization (Tomita & Castillo 2012). By contrast, because postsynaptic *Drosophila* NMJ iGluRs do not function in the absence of Neto, it is not possible to determine whether Neto influences receptor gating, but it is notable that the deactivation of postsynaptic *Drosophila* NMJ iGluRs is much faster than that of their vertebrate kainate receptor Neto complex counterparts. Similar to vertebrate Ca^{2+} -permeable AMPA and kainate receptors (Bowie & Mayer 1995; Kamboj et al. 1995), we previously reported that recombinant *Drosophila* NMJ iGluRs expressed in *Xenopus* oocytes show biphasic rectification and are blocked by spermine and polyamine toxins (Han et al. 2015). In the present study, we found that PhTx, a polyamine toxin derived from wasp venom (Bähring & Mayer 1998; Eldefrawi et al. 1988), blocks both type-A and type-B receptors (Fig. 7). However, voltage-dependent blockade by spermine for vertebrate kainate receptors is reduced in the presence of Neto1 and Neto2 (Brown et al. 2016); by contrast, spermine produces strong voltage-dependent blockade of type-A *Drosophila* NMJ iGluR/Neto complexes (Han et al. 2015) resembling that for vertebrate kainate receptors in the absence of Neto.

Our Neto structure/function studies indicate that a 'minimal Neto' with no intracellular domain can enable the function of type-A receptor *in vitro* and suggest that Neto- α cytosolic domain enhances both deactivation and desensitization (Fig. 1). By contrast, the Neto- β intracellular domain does not appear to modulate receptor gating kinetics. Instead, our previous studies indicate that the Neto- β cytosolic domain functions as a dynamic postsynaptic scaffold critical for PSD assembly and maintenance (Ramos et al. 2015). Blockade of *Drosophila* NMJ iGluRs by PhTx has particular relevance for studies of synaptic plasticity at the NMJ. At the larval NMJ, manipulations that reduce postsynaptic iGluR activity trigger a compensatory increase in neurotransmitter release (DiAntonio et al. 1999; Frank et al. 2006). This form of plasticity has been intensely studied using two experimental paradigms: (1) a chronic (developmental) response in *GluRIIA*^{null} mutants and (2) an acute response

to PhTx (20 μM) applied to dissected larval fillets. The two settings appear to elicit genetically distinct responses (Frank, 2014). Our findings show that external application of PhTx blocks not only type-A (>98%), but also type-B receptors (by ~80%), indicating that both type-A and type-B receptors are impaired by PhTx application during acute potentiation. Thus, the two different plasticity paradigms differ not only in the time frame, developmental vs. acute, but also in the nature of receptors affected; specifically, animals must compensate for the absence of type-A receptors during the developmental paradigm, and for the reduction of both type-A and type-B activities during the acute response, triggered by PhTx application. Because genetically distinct mechanisms recruit type-A and type-B receptors at synaptic sites (Liebl & Featherstone 2008; Parnas et al. 2001; Petzoldt et al. 2014; Ramos et al. 2015; Sulkowski et al. 2016), selective disruption of their function should initiate distinct compensatory mechanisms.

In our experiments, we found that the deactivation kinetics for recombinant *Drosophila* NMJ iGluRs were very fast for both type-A and type-B receptor complexes with Neto- α , τ_{off} 0.64 and 0.42 ms, respectively. Slightly less rapid deactivation was recorded for type-A and type-B receptor complexes with Neto- β , τ_{off} 0.90 and 0.58 ms, respectively. For outside-out patches obtained from larvae with genetically controlled receptor composition, deactivation was two-fold slower compared to recombinant iGluRs for both type-A and type-B receptor complexes: τ_{off} 1.57 ms for A/ α and 1.60 ms for A/ β , and 0.97 ms for B/ β . The different kinetics of deactivation and desensitization for recombinant receptors and native extrajunctional channels observed in our experiments may reflect differences in post-translational modifications or lipid microenvironments surrounding these channels in HEK cells vs. larval muscle. Alternatively, as yet unidentified additional accessory subunits or trans synaptic proteins might modulate receptor function.

Single-channel activity

Prior studies on native *Drosophila* NMJ iGluRs have largely focused on measurements of single-channel conductance (Broadie & Bate 1993a; DiAntonio et al. 1999; Heckmann & Dudel 1997; Nishikawa & Kidokoro 1995). Among these studies, there is good agreement with the results obtained in the present experiments, which reveal a single-channel conductance of 160–170 pS, with no difference between native and recombinant type-A and type-B receptors or Neto isoforms. Our experiments on recombinant *Drosophila* NMJ iGluRs revealed frequent transitions to subconductance states of 75% and 50% of the main state when activated by 10 mM

Table 3. Analysis of single-channel kinetics for recombinant type A/ α and A/ β receptor complexes for control patches, and patches obtained from HEK cells treated with Con A

Recombinant iGluRs	Closed time (ms)	Open time (ms)	Burst length
A/ α	Bimodal	0.74, 3.75	0.76, 3.12
A/ α + Con A	0.6, 3.8	0.67, 7.53	0.64, 11.53
A/ β	Bimodal	1.11, 4.51	1.28, 4.38
A/ β + Con A	Bimodal	5.60, >100 ^a	3.32, >100 ^b

Closed time histograms revealed bimodal distributions for all conditions, but except for A/ α + Con A, the number of events was too small to allow estimate of the lifetime. Log binned open time and Burst Length distributions were fit with the sum of two exponentials as shown in Fig. 3. For receptor complexes with Con A, we observed single openings and bursts of openings that exceeded the length of the 100 ms application of glutamate.

^a Openings of duration in excess of 100 ms were observed.

^b Bursts of duration in excess of 100 ms were observed.

Table 4. Analysis of single-channel kinetics reported in prior studies on native extrajunctional receptors expressed in *Drosophila* muscle

Native iGluRs	Closed time (ms)	Open time (ms)	Burst length
Nishikawa & Kidokoro (1995)	Tau crit 1 ms	NR	0.069, 0.9, 4.9
Broadie & Bate (1993b)	NR	0.23, 2.1	NR
Heckmann & Dudel (1995)	0.073, 0.661	0.071, 1.81	NR

NR indicates not reported.

glutamate. This was observed for both control patches and for patches from cells pretreated with Con A, but it was easier to detect in the latter condition because of reduced desensitization. The presence of subconductance states is a widely observed feature of the gating of vertebrate AMPA and kainate receptors (Baranovic et al. 2022; Coombs & Cull-Candy, 2021; Daniels et al. 2013; Rosenmund et al. 1998) and it is surprising that this has not been reported before for native extrajunctional *Drosophila* NMJ iGluRs. Inspection of published raw data from prior single-channel recording experiments reveals hints of substate activity, although it is possible that this results from brief transitions between fully open and closed states that were not resolved (DiAntonio et al. 1999; Heckmann & Dudel 1995; Heckmann & Dudel 1997; Nishikawa & Kidokoro 1995).

Because, for measurement of single-channel lifetimes, we used a cutoff value of 200 μ s to generate the list of idealized events, and a critical time of 2 ms for burst length analysis, we did not resolve brief events with micro-second lifetimes as reported in prior studies on native receptors (Chang & Kidokoro 1996; Heckmann & Dudel 1995). Because of the limited number of single-channel events recorded in control patches as a result of rapid desensitization, combined with their short duration, we did not calculate closed time distributions, and we limited our analysis to recombinant type-A receptors for which openings were well resolved. For control patches, the open

time distributions were comparable to those reported for wild-type extrajunctional receptors, as summarized in Tables 3 and 4 (Broadie & Bate 1993b; Heckmann & Dudel 1995; Nishikawa & Kidokoro 1995), whereas, for patches from HEK cells treated with Con A, there was a substantial increase in open time. For the burst length distribution, there is only a single observation in the literature for native iGluRs, with medium and long-duration events of lifetime 0.9 and 4.9 ms (Nishikawa & Kidokoro 1995) (Table 4); these values are comparable to those obtained for recombinant receptors A/ α , 0.76 and 3.12 ms, and A/ β , 1.28 and 4.38 ms (Table 1). Overall, the treatment with Con A increased the lifetime of long-duration openings for A/ α by almost four-fold; for A/ β , the lifetime of long-duration events in the open time and burst duration distributions exceeded 100 ms, resulting in tail currents following removal of glutamate (Figs 3C and 4D, F and Table 3). In addition, we analysed the open time distribution of substates within bursts and found additional subtype specific differences (Fig. 5 and Table 3).

The decay of synaptic currents at the NMJ and central synapses

Building on the pioneering discovery by Fatt & Katz (1952) of quantal synaptic transmission, a central tenet of synaptic physiology at the vertebrate NMJ is that

the rate of closure of ion channels activated by acetylcholine determines the time course of the synaptic response, for which decay of the endplate current was shown to be determined by the intrinsic rate constant for closure of nicotinic acetylcholine receptors (Magleby & Stevens 1972). At central synapses, which, similar to the *Drosophila* NMJ, use glutamate as a neurotransmitter, the situation is more complex, and the EPSC kinetics vary throughout the brain (Jonas, 2000). In the neocortex, the mEPSC decay time constant, τ_{mEPSC} 2.3 ms, is only slightly slower than time constant for channel closure, τ_{off} 2.1 ms (Hestrin 1992), but, for hippocampal neurons synaptic currents have slower kinetics, τ_{mEPSC} 4–8 ms (Hestrin et al. 1990). By contrast, sub-millisecond mEPSC decay and iGluR deactivation have been recorded from neurons in brain stem auditory pathways, interneurons and cerebellar Purkinje cells, probably reflecting differences in the subunit and auxiliary proteins present in different iGluR complexes (Barbour et al. 1994; Geiger et al. 1995; Raman et al. 1994).

In our experiments, we found that the decay time constants of mEJCs, recorded from larvae with genetically controlled receptor and Neto splice variant composition, were three to five times slower than for deactivation measured for native extrajunctional receptor-Neto complexes of the same composition. By contrast, the rate of decay mEJC was almost identical to the rate of desensitization for A/ α , τ_{mEJC} 4.2 ms, τ_w 4.6 ms; for A/ β , the rate of desensitization was 1.4-fold slower than the decay of the synaptic current, τ_{mEJC} 6.5 ms, τ_w 9.4 ms; unexpectedly, for B/ β , the rate of desensitization was 1.8-fold faster than the synaptic current, τ_{mEJC} 4.9 ms and τ_w 2.7 ms. Because animals with B/ α receptors die during late embryogenesis and are unable to hatch, B/ α receptors probably deactivate and desensitize extremely rapidly and cannot sustain normal synaptic transmission and muscle contraction. Overall, our results suggest that desensitization plays a major role in determining the kinetics of synaptic transmission at the larval NMJ. We are not aware of any reports for which the synaptic current decay at vertebrate synapses is determined exclusively by the rate of desensitization. However, for some vertebrate CNS glutamatergic synapses, desensitization has been shown to contribute to a slow component of synaptic currents (Barbour et al. 1994; Koike-Tani et al. 2005). In addition, desensitization shapes the response to multiquantal neurotransmitter release, and to paired pulse stimulation at vertebrate glutamatergic synapses with fast kinetics (Trussell et al. 1993).

In our experiments, mEJCs were recorded using two-electrode voltage clamp which revealed decay time constants of 4–6 ms, which varied with subunit composition. A survey of the literature reporting *Drosophila* NMJ mEJCs recorded using extracellular focal recording (Cooper et al. 1995; Dawson-Scully et al.

2007; Heckmann & Dudel 1998; Karunanithi et al. 2018; Stewart et al. 1994) revealed values comparable to those recorded in the present experiments, suggesting that the slow kinetics recorded using two-electrode voltage clamp from *Drosophila* larval muscle are not artefacts arising from poor space clamp. In summary, our results suggest that the time course of decay of the concentration of glutamate in the synaptic cleft of the larval NMJ is slow compared to conventional synapses. Such slow glutamate clearance may reflect the complex regulation of extracellular glutamate concentration at larval NMJ: First, glutamate is cleared by EAAT1, encoded by the sole *EAAT* gene within the fly genome, which is expressed in larval glia (Nguyen et al., 2024) but at much lower levels than in the adult periphery (Rival et al. 2006). Second, the glial xCT transporter, Genderblind (Gb), functions during larval stages to increase the extracellular glutamate concentration and thus limit the synaptic recruitment of NMJ iGluRs (Augustin et al. 2007); in the absence of Gb, extracellular glutamate is reduced by half and the synaptic accumulation of postsynaptic glutamate receptors is increased by two- to three-fold.

The fly NMJ is a powerful genetic model for investigating conserved mechanisms for synapse assembly, development and homeostasis. Unlike in vertebrates, where *in vivo* studies of kainate receptors have been technically very challenging, disruption of kainate receptor biology at the fly NMJ triggers a wide range of morphological and behavioural phenotypes, from a complete loss of postsynaptic receptors and synaptic structures, which causes embryonic paralysis and developmental lethality, to defects in the assembly and maintenance of PSDs, which generally induce locomotor deficits, and finally to subtle changes in postsynaptic composition and impairments in synaptic plasticity. Within this wide range of biological outcomes, investigations of fly NMJ iGluRs and their modulation by Neto have the potential to reveal new functions for iGluRs and Neto and new modalities of regulation. Our study paves the way to parse out and elucidate the multiple functions and regulation of *Drosophila* kainate receptors and their auxiliary proteins.

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Additional information

Data availability statement

All data supporting the results of the present study are included within the published paper.

Competing interests

The authors declare that they have no competing interests.

Author contributions

This work was performed in the Serpe laboratory at the NICHD, NIH. T.H.H., M.L.M. and M.S. designed the work. T.H.H. and C.I.R. generated the data. T.H.H., R.V., M.L.M. and M.S. performed the analysis. All authors contributed to drafting and revising the manuscript, approved the final version of the manuscript submitted for publication, and agreed to be accountable for all aspect of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All persons designated as authors qualify for authorship and all those who qualify for authorship are listed.

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Keywords

Drosophila iGluRs, gating properties, Neto, neuromuscular junction

Supporting information

Additional supporting information can be found online in the Supporting Information section at the end of the HTML view of the article. Supporting information files available:

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